

PyGPLA: Generalized Phase Locking Analysis in Python

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Abstract

Concise motivation, methods, results, and conclusions in one paragraph. Keep under ~200 words.

Summary

Briefly explain why multichannel spike-field coupling is hard to interpret and how GPLA provides a dominant-coupling summary (gPLV, LFP/spike vectors) with statistical guarantees.

Statement of need

- Why existing pairwise PLV/phase metrics are insufficient for large-channel recordings.
- Gap: accessible, Python-native GPLA with stats, whitening, and simulations.

- Who benefits: systems neuroscience labs analyzing LFP + spikes at scale.

Package overview

- High-level API: `pygpla.api.gpla` returning gPLV, vectors, stats.
- Core math: coupling matrix + SVD factorization.
- Statistics: RMT analytic test and spike-jitter surrogates.
- Preprocessing: whitening, normalization, same-electrode correction.
- Simulations: phase-locked and transient coupling generators.

Methods

Data model and preprocessing

Describe spike trains (units $\times$ samples $\times$ trials) and analytic LFPs (channels $\times$ samples $\times$ trials), unit selection, whitening options, and normalization. Note thresholds for spike counts and same-electrode handling.

Coupling matrix and GPLA decomposition

Define the complex coupling matrix (C), computed from analytic LFPs and spike trains, and its SVD:

$$C = U\Sigma V^H, \quad \text{gPLV} = \sigma_1.$$

Describe column normalization, phase rotation (set mean LFP phase to zero), and optional unwhitening.

Statistical testing

Explain the RMT-based analytical test (Marchenko–Pastur bound on (χ^2_1)) and surrogate assessments using spike-jitter variants. Include parameters: number of surrogates, jitter window, alpha, jitter type.

Simulations and datasets

List synthetic datasets (von Mises phase-locked spikes, transient locking) and any real recordings (e.g., Utah array). Document parameter ranges, sampling rates, preprocessing filters, and seeds.

Results

Summaries for figures/tables once added: simulation benchmarks (detection power/Type I error), robustness to dimensionality, and a real-data case study highlighting LFP/spike vector interpretation.

Usage

- Installation: `pip install .[sim,docs]` (or editable mode during development).
- Minimal example: load/generate data → call `gpla` → inspect `gPLV`, vectors, stats.
- Reproducing figures: list script names, commands, and seeds.

Availability

Repository URL, version tag, license, and issue tracker. Note dependencies (NumPy, SciPy for sims, pytest for tests, Quarto for this manuscript).

Acknowledgments

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References

Add a `references.bib` alongside this file; cite inline, e.g., `[@safavi2023gpla]`, and Quarto will format automatically.